

photon energy (log scale)



inside a cavity

THz / far-IR

H-bond network

1-40 meV

10-300 cm^{-1}

THz-Raman, 2D-THz

Mid-IR / Raman

O-H stretch, bend

0.12-0.5 eV

1000-4000 cm^{-1}

IR, Raman, SFG

UV

photodissociation

7-11 eV

110-180 nm

abs., fragments

X-ray

O 1s core level

530-560 eV

O K-edge

XAS, RIXS, TR-XAS

photons no longer a weak perturbation · Rabi splitting 10-500 meV